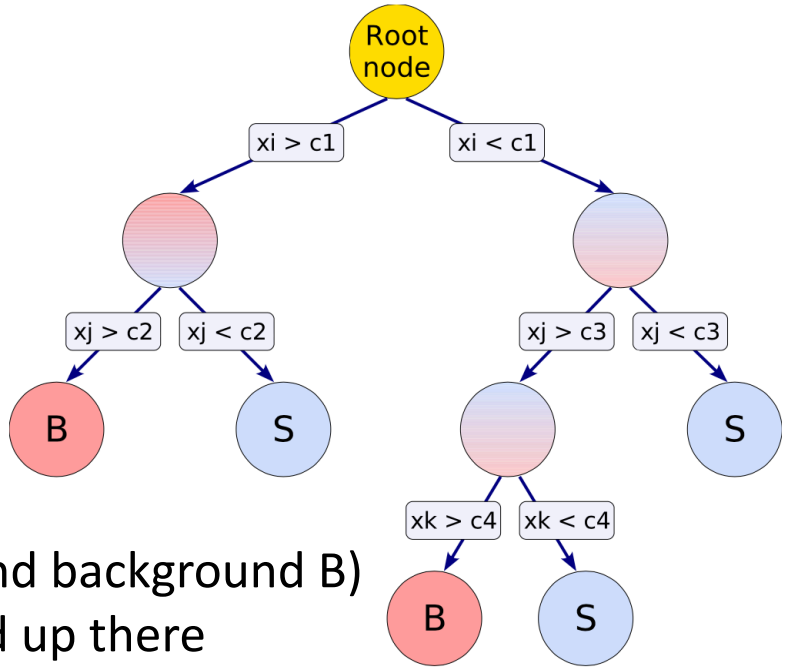
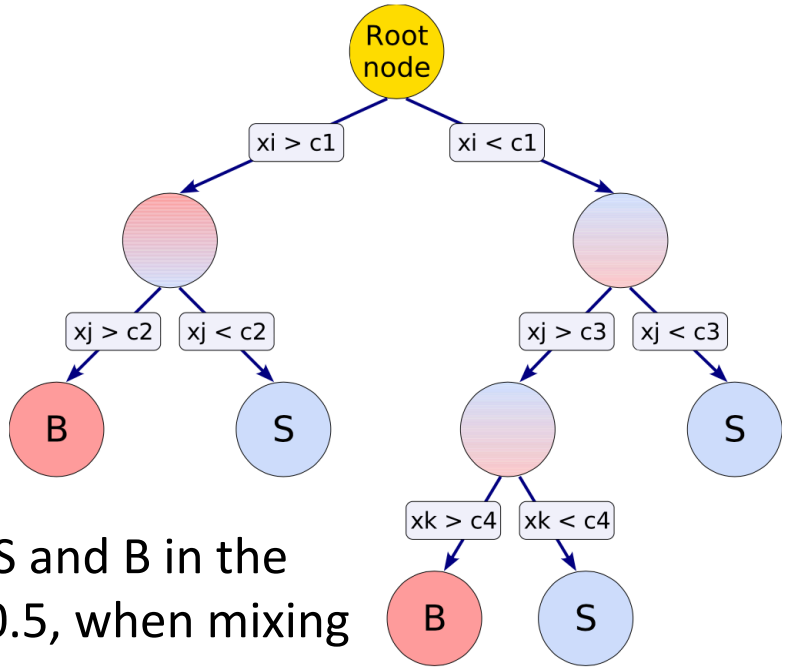
- **Boosted decision trees** are a **multivariate classification technique** that builds on the concept of simple **decision trees**
- A **decision tree** is a tree-like structure consisting of a series of binary splits that slice the sample up based on a series of discriminating input variables



Leaves are given **class labels** (e.g. signal S and background B) depending on how many S and B events end up there ($purity = S/(S+B)$ above/below threshold)

A **branch** is the sequence of binary splits that lead from the root node to a specific leaf

- During training, the decision tree finds, at each **node**, the discriminating variable and the cut that give the best separation by minimising a figure of merit
- Possible choices of figures of merit are *misclassification error, cross-entropy, Gini index*, etc.



They all estimate the degree of "mixing" of S and B in the two split samples, and have a maximum at 0.5, when mixing is maximal

E.g. **misclassification error** $= 1 - \max(p, 1 - p)$ where $p = \frac{S}{S+B}$.

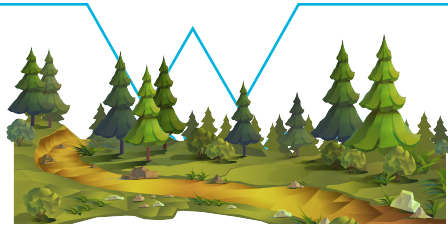
Purity can be **weighted:** $p = \frac{\sum_{i \in Signal} w_S^i}{\sum_{i \in S} w_S^i + \sum_{i \in B} w_B^i}$

Problem: single decision trees are very sensitive to statistical fluctuations!

- (Important) good practice: use different samples for different steps
  - **Training**
  - **Validation** (sample on which to assess the performance of the DT, while tuning its hyper-parameters)
  - **Testing** (unbiased evaluation of final performance)

- …But can go beyond that!

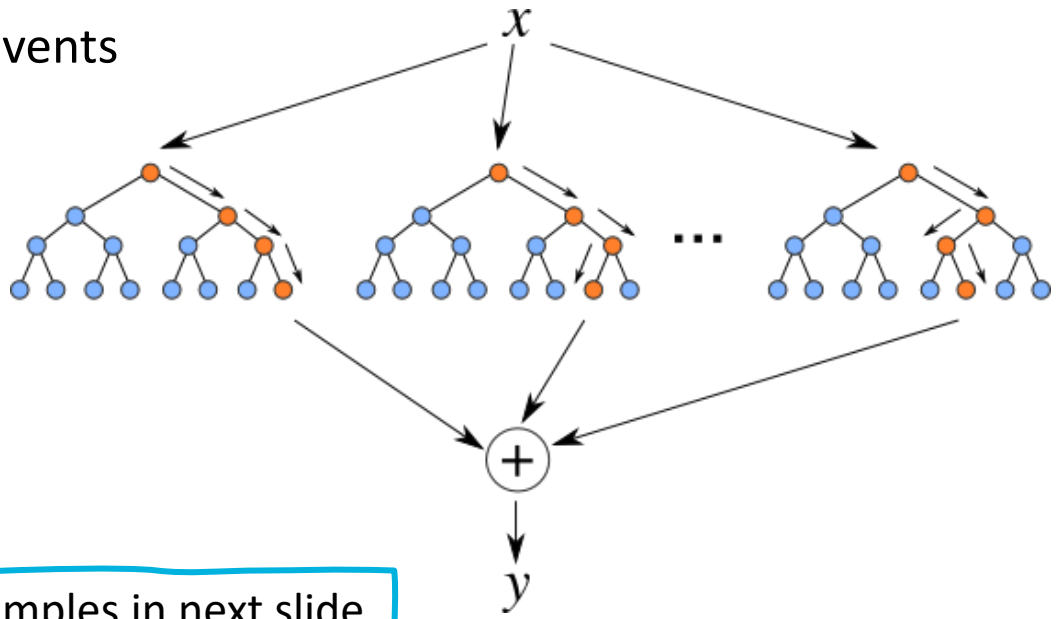
…solution:



Boosting: going from a tree to a forest!

Many trees are derived from the same training sample by **reweighting** events

When using the BDT for classification, for each event, the outputs of all trees are combined by a weighted average of the individual decision trees scores



Many possible strategies! Some examples in next slide

• Adaptive boosting

In the calculation of the figure that is minimised, e.g. misclassification error, each of the events is multiplied by a weight.

Events that were misclassified during the training of a decision tree are given a higher event weight in the training of the following tree

Recipe

- Derive weight of whole tree α_K based on misclassification rate and boost factor
- Reweight events: $w_i^K \rightarrow w_i^K \times e^{\alpha_K}$
- Calculate tree output

$$T(i) = \frac{1}{\sum_{k=1}^{N_{tree}} \alpha_K} \sum_{k=1}^{N_{tree}} \alpha_K T_K(i)$$

K-th tree output for event i

BDT output for event i

K-th tree weight (can depend on a boosting factor and misclassification rate)

For each event, the output of single decision tree is a yes/no (or 0/1) variable; output of BDT is a continuous variable!

At the end, can calculate a BDT **score** as an average accuracy: how often is the BDT correct?

Bagging

- Each tree is trained using a sub-sample of the entire training sample
- The BDT response is stable against statistical fluctuations → higher performance

Some references

- ROOT TMVA manual
- https://indico.cern.ch/event/472305/contributions/1982360/attachments/1224979/1792797/ESIPAP_MVA160208-BDT.pdf
- https://scikit-learn.org/stable/modules/generated/sklearn.tree.DecisionTreeClassifier.html